
Congestion-Aware Exit Choice: Decision Timing in a Worked Simulation

CSF Toolchain — reproducible worked example

Statistics, figures and tables generated from tracked per-seed simulation records

Abstract

This worked example connects an implemented multi-agent simulator to reproducible quantitative figures. In a synthetic hall with 400 agents and three exits, a nearest-exit rule produces a mean clearance time of 426.66 ± 5.98 s, whereas a distance-plus-queue rule applied at every movement step produces 184.54 ± 15.87 s (mean $\pm$ sample standard deviation across five runs). The continuous capacity reference is 152.21 s; the dynamic mean is 1.21 times this reference. We separate decision timing from the empty-queue equivalence of three initialization rules, show all saved seed outcomes, and report uncertainty in every quantitative panel. These descriptive results concern the supplied synthetic model; they establish neither global optimality nor safety in a real venue.

1 Problem and scope

Agents must choose one of three exits in a rectangular hall. The purpose is to illustrate a transparent experiment-to-figure workflow, rather than to calibrate pedestrian behavior or train a reinforcement-learning policy. Exit widths differ, but all agents in the main experiment have the same movement speed. The central question is how clearance time changes when moving agents can repeatedly update their exit choice using the current queues.

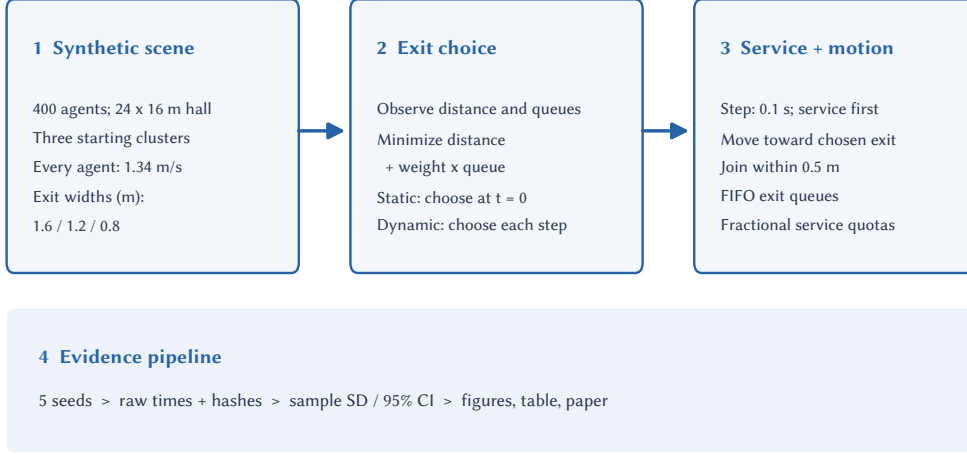
The simulator and saved experiments live in the repository’s original evacuation example code directory. The English example derives its published numbers from the saved results. No external empirical dataset, collision model, social-force dynamics, learned policy, or population-level safety validation is included. Figure 1 describes the actual implemented steps.

2 Model and implementation

The hall is 24×16 m. Exit coordinates are (0, 8), (24, 8) and (12, 0), with widths 1.6, 1.2 and 0.8 m. Multiplying these widths by the specified synthetic flow coefficient 0.73 persons/(m s) gives service rates $\mu = (1.168, 0.876, 0.584)$ persons/s. Agents move at 1.34 m/s. Initial positions

A congestion-aware exit-choice simulation

Implemented rules, service timing, and traceable evidence



Synthetic model; no collisions, learned policy, or real-world calibration.

Figure 1. Implemented simulation and evidence pipeline. Every moving agent observes distances and queue lengths. A fixed-step service rule and FIFO queues determine completion.

are sampled from three clipped Gaussian clusters centered at (12, 3), (19, 8) and (5, 8), with proportions 0.60, 0.25 and 0.15. Standard deviations are 2.2 and 1.6 m in the two coordinate directions. Coordinates are clipped to a 0.30 m inset from the hall boundary.

State consists of agent positions, assigned exits, arrived/departed flags, FIFO queues, cumulative exit counts and fractional service credits. At each 0.1 s step, exits first accumulate $\mu_e \Delta t$ credits and consume the integer portion. Unused integer slots are discarded; only the fractional remainder persists. Completion is checked, then distances and queues are observed, moving agents select exits, movement is applied, and agents within 0.5 m join their assigned queues. Queued agents do not move or reselect an exit under the dynamic rule.

The congestion-aware heuristic selects

$$e_i(t) = \arg \min_e \{d_{ie}(t) + \lambda Q_e(t)\},$$

where distances are in meters, queue lengths are in persons and λ is in m/person. The main comparison uses $\lambda = 3$. This is a heuristic distance penalty, not a derived marginal externality tax or an explicit proportional-capacity assignment algorithm. Ties use the exit order in the code.

Initialization equivalence. At initialization, every queue is empty. Hence distance-plus-queue reduces to nearest-exit selection. The initialization-only shortest-queue implementation also breaks ties by distance. These three implementations therefore produce the same assignments and saved completion times in the main experiment. They are presented as one baseline, rather than counted as three independent comparisons. The arrival-time variant is a separate policy and remains a separate table row.

Capacity reference and time quantization. Ignoring travel and idle service slots gives the continuous reference

$$T_{\text{ref}} = \frac{N}{\sum_e \mu_e} = 152.21 \text{ s.}$$

The implementation credits service at time zero. By recorded time T , the total number of possible service slots is at most $\sum_e \mu_e (T + \Delta t)$, so a compatible discrete-time lower bound is $T \geq T_{\text{ref}} - \Delta t$.

We label T_{ref} as a continuous capacity reference in the figures. Neither reference is an attainable optimum for the full queue-and-motion problem, and an observed residual cannot be uniquely attributed to travel without additional experiments.

3 Experiments and statistical reporting

The saved main and sweep experiments use seed identifiers 2026–2030. Each reported setting contains five complete clearance times. We recompute the mean and sample standard deviation from these raw values, rather than copying rounded summaries. Confidence intervals for the mean use the two-sided Student- t interval with four degrees of freedom. Dots in Figure 2 show the individual saved outcomes; intervals describe uncertainty in the mean, not the dispersion of individual trajectories. With five synthetic runs, the normal-mean interval is an assumption-dependent descriptive summary.

Main policies are random assignment at initialization, nearest exit at initialization, shortest queue on arrival, and the dynamic distance-plus-queue heuristic. Sweep experiments vary congestion weight, the proportion of agents using the dynamic rule, and population size. The dynamic-share experiment varies the eligible share of agents, not the time interval between decisions. No significance test, multiplicity correction, independently tuned test set, or proof of global optimality is provided. Shared seed identifiers alone do not establish common random events for all policy-dependent random draws; we do not report a paired confidence interval for policy differences.

Policy	Mean (s)	Sample SD (s)	95% CI of mean (s)	Mean / ref.
Random once	223.22	15.55	203.91 to 242.53	1.47
Nearest once (3 equivalent rules)	426.66	5.98	419.24 to 434.08	2.80
Shortest queue on arrival	426.66	5.98	419.24 to 434.08	2.80
Dynamic distance + queue	184.54	15.87	164.84 to 204.24	1.21

Table 1. Main comparison, recomputed from five complete raw runs per policy. SD is the sample standard deviation. Mean/ref. uses the continuous capacity reference.

4 Results and sensitivity

Table 1 reports the main comparison. The dynamic heuristic has a lower observed mean than nearest-exit initialization, and its mean/reference ratio is 1.21. This is a descriptive comparison within the model. It does not imply that the heuristic achieves a given percentage of an optimal solution or that its residual has a single causal explanation.

For the congestion-weight sweep, only the sampled weights {3, 8} fall within 3% of the smallest observed mean. This is a discrete set, not a continuous plateau: a tested weight between those points can lie outside the threshold. The selected weight is a post-hoc minimum over the saved sweep and is not evidence of a unique optimal parameter.

Increasing the dynamic share reduces the observed means in the saved experiment. Both main policies have larger clearance times as population grows, and their descriptive separation persists at the four tested sizes. Every sweep panel retains its available uncertainty. These comparisons cannot establish which omitted behavior or resource constraint would dominate outside the supplied parameter range.

Decision timing changes clearance time

5 synthetic runs | dots: raw seeds | intervals: mean 95% CI

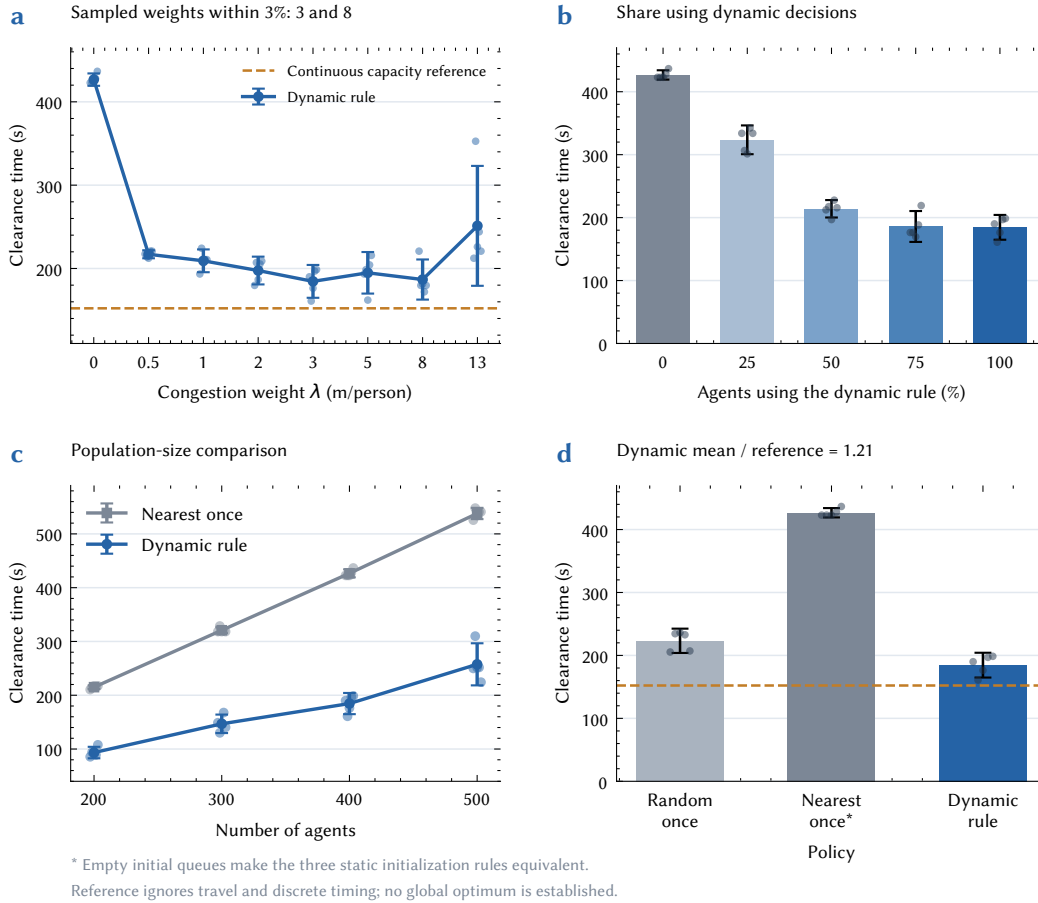


Figure 2. Quantitative evidence from saved synthetic experiments. (a) All tested congestion weights, including zero. (b) Share of agents applying the dynamic rule. (c) Instance-size comparison. (d) Main policies. Dots are raw runs; error bars are 95% Student-*t* intervals of the mean. The dashed reference ignores travel and discrete service timing.

5 Verification, limitations and reproduction

The numerical generator checks seed coverage, successful completion flags, positive finite clearance times, consistent main/sweep configurations and saved completion-time equivalence of the three initialization rules. It records SHA-256 hashes of both input result files. These checks verify the publication data path; they do not independently audit every trajectory or validate the model against observations. The repository's separate `validated-dispatch` example demonstrates an independent capacity and conservation audit for a simpler scheduling model.

Remaining limitations include the absence of physical contact, heterogeneous speeds in the main experiment, empirical calibration, independently audited historical trajectories and larger replication budgets. The recorded reassignment counter for the dynamic policy is not a count of every movement-step choice; it is therefore omitted from the comparison. Rounded average exit counts are not treated as exact conserved totals. Gini is not used as an objective because equal exit counts need not reflect unequal service capacities.

From the repository root, run `python examples/evacuation-en/make_example.py`. It regenerates the tracked statistics, both figures in PDF/SVG/PNG, the comparison table and `results/numbers.tex`. The paper inputs the generated macros and table. The source-statistics generator's `--check` mode fails if the published JSON is stale. Figure exports suppress variable date metadata and use stable SVG identifiers; repeatability is checked within the same installed rendering environment. Fonts and library versions can change geometry across machines.

Compile this paper with `csf_build.py --tex examples/evacuation-en/paper.tex`, then render and inspect all PDF pages. This short example is an internal reproducibility demonstration, not a claim that the full competition manuscript gates or AnyMath deployment tests have passed.